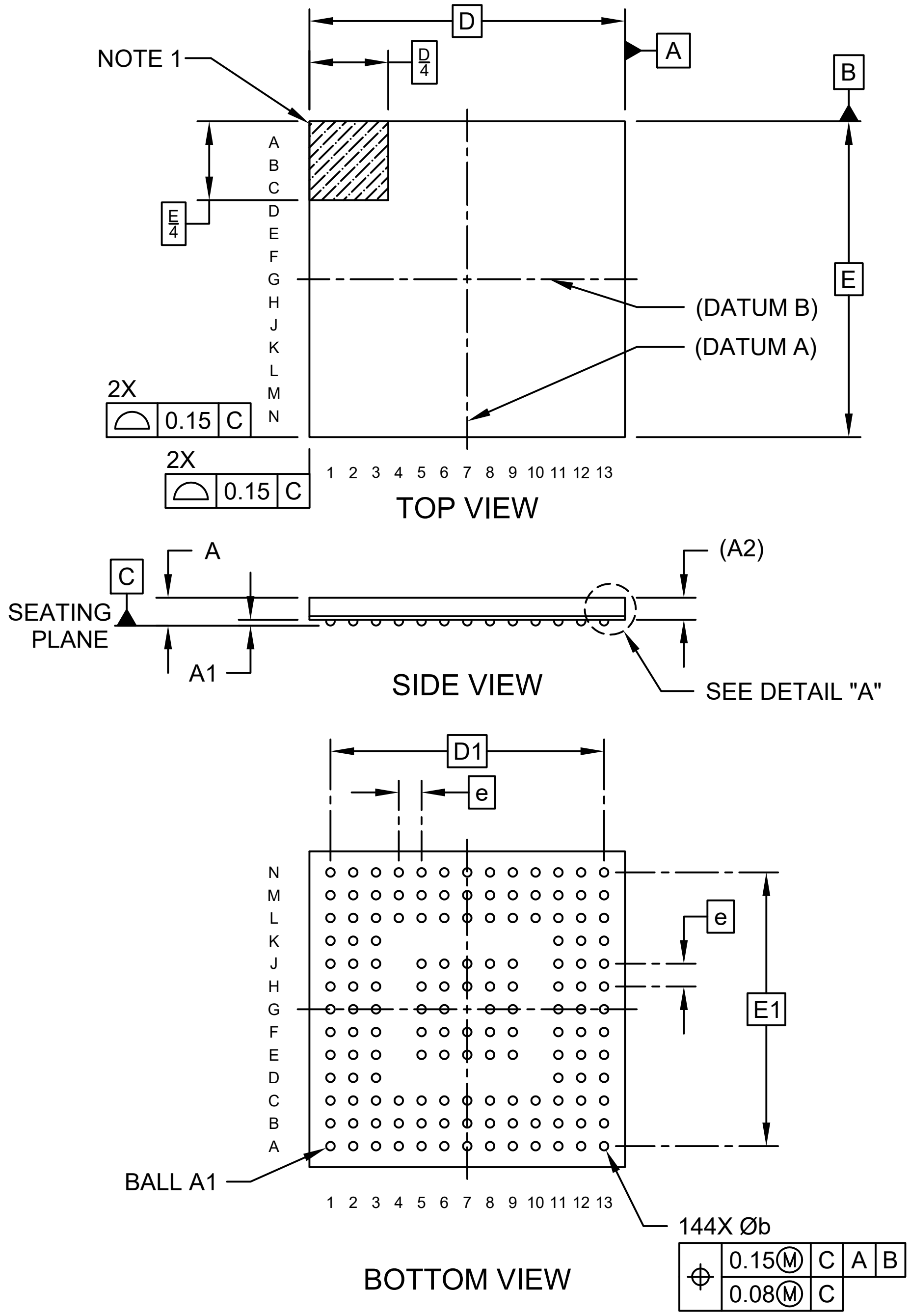
## 144-Ball Very, Very Thin Fine Pitch Ball Grid Array (SZ) -9x9x0.8 mm Body [WFBGA]

| | |
|---|---|
| **Note:** | For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging |



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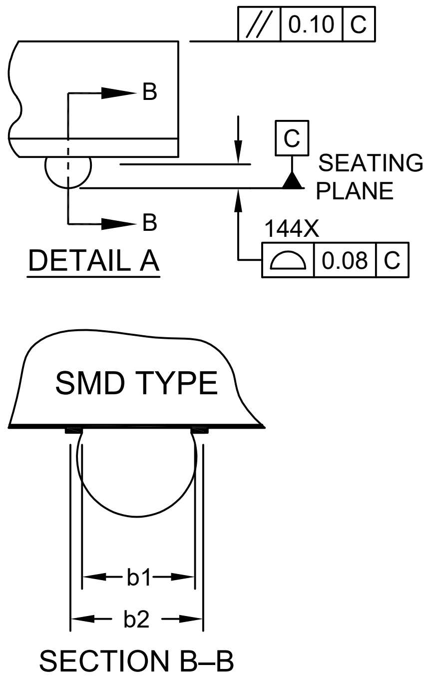


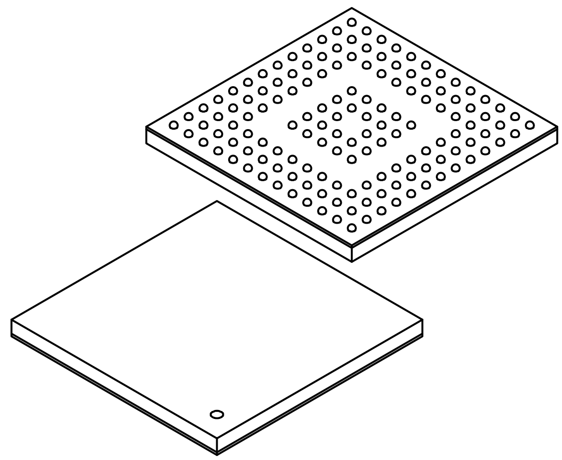

| Units | | MILLIMETERS | | |
|---|---|---|---|---|
| Dimension Limits | | MIN | NOM | MAX |
| Ball Pitch | e | | 0.65 BSC | |
| Overall Height (see Note 7) | A | - | 0.70 | 0.80 |
| Standoff | A1 | 0.12 | 0.17 | 0.22 |
| Terminal Thickness | A2 | | 0.53 REF | |
| Overall Length | D | | 9.00 BSC | |
| Overall Ball Pitch | D1 | | 7.80 BSC | |
| Overall Width | E | | 9.00 BSC | |
| Overall Ball Pitch | E1 | | 7.80 BSC | |
| Ball Diameter | b | 0.20 | 0.25 | 0.30 |
| Finished Solder Mask Opening | b1 | 0.22 | 0.25 | 0.28 |
| Finished Bottom Ball Pad | b2 | 0.30 | 0.35 | 0.40 |

Notes:

1. Ball A1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.
4. Primary Datum "C" and Seating Plane are defined by the spherical crowns of the contact solder balls.
5. Dimension "A" does not include attached external featues, such as heat sink or chip capacitors.
6. The package ball solderable surface is solder-mask defined (SMD) type.
7. Maximum Overall Height for LAN7515 parts only is 0.90 mm.

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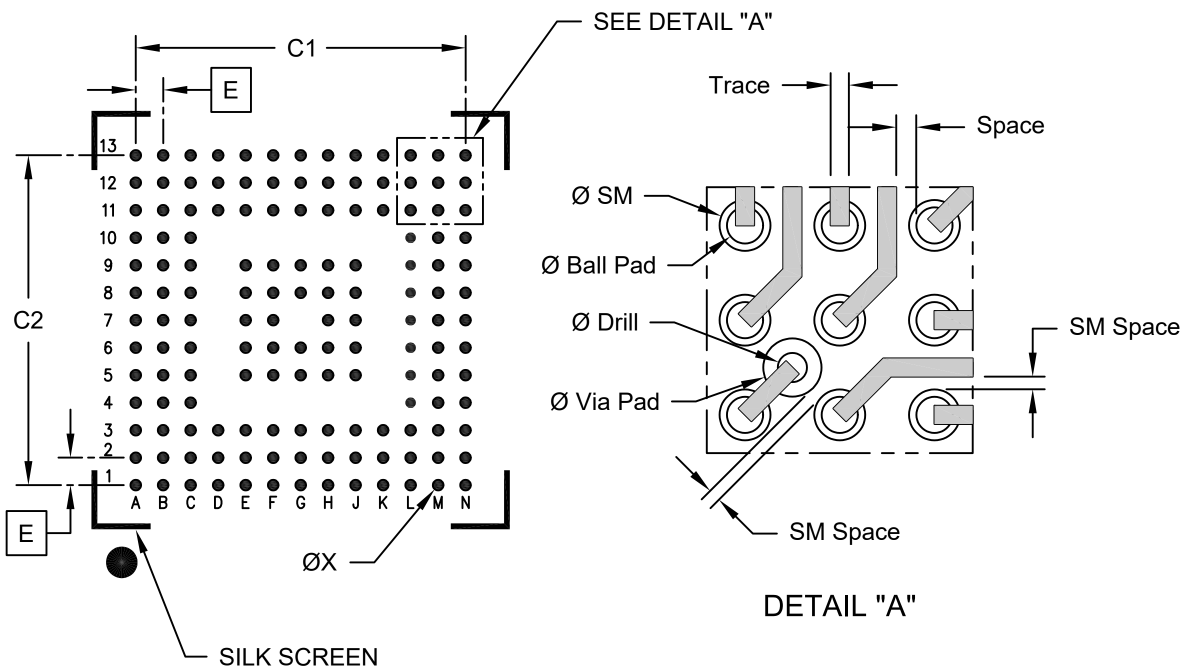


RECOMMENDED LAND PATTERN

| | Units | MILLIMETERS | | |
|---|---|---|---|---|
| | Dimension Limits | MIN | NOM | MAX |
| Contact Pitch | E | | 0.65 BSC | |
| Overall Contact Pitch | C1 | | 7.80 | |
| Overall Contact Pitch | C2 | | 7.80 | |
| Contact Pad Diameter | X | | 0.25 | |

Routing Dimensions

| Units | mm |
|---|---|
| Feature | |
| Ø PAD | 0.250 |
| Ø SM | 0.350 |
| Trace Width | 0.125 |
| Space (Min.) | 0.135 |
| SM Space (Min.) | 0.085 |
| Ø Via Pad | 0.400 |
| Ø Drill | 0.200 |

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.

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